

A Direct AnyPrice-Share Certificate for a $39/40$ Obstruction

June 24, 2026

Abstract

We examine the open problem of determining the optimal universal approximation constant for AnyPrice-share allocations of indivisible goods with additive valuations and arbitrary entitlements. The result below is not a solution of the open problem. It gives a direct, checkable certificate for the upper-bound side: there is a three-agent equal-entitlement instance in which every agent has AnyPrice share exactly 40, but every allocation gives some agent value at most 39. Hence no universal approximation factor strictly larger than $39/40$ is possible. Together with the theorem of Babaioff, Ezra, and Feige giving a $3/5$ approximation for additive valuations with arbitrary entitlements, this certifies the interval

$$\frac{3}{5} \leq \alpha^* \leq \frac{39}{40}.$$

Thus the note should be read as partial progress, or more precisely as a direct APS-level certificate for a known obstruction, rather than as a determination of α^* .

1 Introduction and status of the result

For indivisible goods, Babaioff, Ezra, and Feige introduced the AnyPrice share, abbreviated APS, as a weighted fairness benchmark for agents with arbitrary entitlements. They proved that for additive valuations one can always give every agent at least a $3/5$ fraction of her APS, and they asked for the optimal universal approximation constant. Let this constant be denoted by α^* .

This note proves the following limited statement.

Theorem 1 (Certified interval). *For additive valuations and arbitrary entitlements,*

$$\frac{3}{5} \leq \alpha^* \leq \frac{39}{40}.$$

The upper bound already holds for three agents with equal entitlements $1/3$.

The lower bound in Theorem 1 is the theorem of Babaioff, Ezra, and Feige. The contribution of this note is the direct upper-bound certificate in Sections 3–5: in the displayed instance, the APS of each of the three agents is exactly 40, while the largest possible minimum utility over all allocations is 39.

Remark 1 (Not a solution). *This does not determine α^* . It leaves open the interval*

$$\frac{3}{5} \leq \alpha^* \leq \frac{39}{40}.$$

A full solution would require either a universal existence theorem with guarantee $39/40$ or better, or a sharper counterexample with ratio below $39/40$, or matching lower and upper bounds at some other value in the interval.

2 Definitions

Let M be a finite set of indivisible goods and let N be a finite set of agents. Agent $i \in N$ has an additive nonnegative valuation

$$v_i(S) = \sum_{g \in S} v_i(g), \quad S \subseteq M,$$

and an entitlement $b_i > 0$, with $\sum_{i \in N} b_i = 1$. A price vector is a vector

$$p \in P := \left\{ p \in \mathbb{R}_{\geq 0}^M : \sum_{g \in M} p_g = 1 \right\}.$$

The AnyPrice share of agent i is

$$\text{APS}_i = \min_{p \in P} \max_{S \subseteq M : \sum_{g \in S} p_g \leq b_i} v_i(S).$$

Equivalently, a price setter chooses item prices of total one, and the agent then chooses her most valuable bundle of total price at most her entitlement.

For an instance, an allocation is a partition $(A_i)_{i \in N}$ of M among the agents. Define α^* to be the largest real number such that every additive instance with arbitrary entitlements admits an allocation satisfying

$$v_i(A_i) \geq \alpha^* \text{APS}_i \quad \text{for all } i \in N.$$

3 The three-agent instance

There are three agents, denoted R, C, U , each with entitlement $1/3$. There are nine goods, arranged only for readability as

$$\begin{array}{ccc} e_1 & e_2 & e_3 \\ e_4 & e_5 & e_6 \\ e_7 & e_8 & e_9. \end{array}$$

The agents' additive valuations are as follows, with entries listed in row-major order:

$$v_R = \begin{pmatrix} 1 & 16 & 23 \\ 26 & 4 & 10 \\ 12 & 19 & 9 \end{pmatrix}, \quad v_C = \begin{pmatrix} 1 & 16 & 22 \\ 26 & 4 & 9 \\ 13 & 20 & 9 \end{pmatrix}, \quad v_U = \begin{pmatrix} 1 & 15 & 23 \\ 25 & 4 & 10 \\ 13 & 20 & 9 \end{pmatrix}.$$

4 The APS values are exactly 40

Lemma 1. *In the instance of Section 3,*

$$\text{APS}_R = \text{APS}_C = \text{APS}_U = 40.$$

Proof. First we prove $\text{APS}_X \geq 40$ for each $X \in \{R, C, U\}$. It is enough to exhibit, for each agent, a partition of the goods into three bundles each of value at least 40. Indeed, under any price vector of total price one, at least one bundle in such a partition has price at most $1/3$; that bundle is affordable for an agent with entitlement $1/3$ and has value at least 40.

For R , use the row partition

$$\{e_1, e_2, e_3\}, \quad \{e_4, e_5, e_6\}, \quad \{e_7, e_8, e_9\}.$$

Each row has R -value 40. For C , use the column partition

$$\{e_1, e_4, e_7\}, \quad \{e_2, e_5, e_8\}, \quad \{e_3, e_6, e_9\}.$$

Each column has C -value 40. For U , use the partition

$$\{e_2, e_4\}, \quad \{e_3, e_5, e_7\}, \quad \{e_1, e_6, e_8, e_9\},$$

whose three U -values are also 40. Hence $\text{APS}_X \geq 40$ for all three agents.

It remains to prove $\text{APS}_X \leq 40$. For each agent X , we exhibit a price vector under which no affordable bundle has value greater than 40. Write $p_j^X = q_j^X / D_X$. The following table gives integer weights q^X summing to D_X :

X	D_X	$q^X = (q_1^X, \dots, q_9^X)$
R	23	(0, 3, 4, 5, 1, 2, 2, 4, 2)
C	66	(1, 9, 12, 14, 2, 5, 7, 11, 5)
U	63	(1, 8, 12, 13, 2, 5, 7, 10, 5)

The exact finite checks for these weights are:

X	$\max\{v_X(S) : 3q^X(S) \leq D_X\}$	$\min\{q^X(S) : v_X(S) \geq 41\}$
R	40	8
C	40	23
U	40	22

Here $q^X(S) = \sum_{e_j \in S} q_j^X$. Since $\sum_j q_j^X = D_X$, the vector $p^X = q^X / D_X$ has total price one. A bundle is affordable for agent X exactly when $q^X(S) / D_X \leq 1/3$, equivalently $3q^X(S) \leq D_X$. The final column is strictly larger than $D_X/3$ in all three cases:

$$8 > 23/3, \quad 23 > 66/3, \quad 22 > 63/3.$$

Therefore no bundle of value at least 41 is affordable under the corresponding price vector. Since all valuations are integral, every affordable bundle has value at most 40. Thus $\text{APS}_X \leq 40$ for all $X \in \{R, C, U\}$.

The two numerical assertions in the last columns of the table are finite checks over the 2^9 possible bundles. Appendix A gives an exact integer checker. $\square$

5 No allocation gives all three agents value 40

Lemma 2. *No allocation of the nine goods gives all three agents value at least 40.*

Proof. For each good, let

$$\mu(e_j) = \max\{v_R(e_j), v_C(e_j), v_U(e_j)\}.$$

Thus

$$\mu = \begin{pmatrix} 1 & 16 & 23 \\ 26 & 4 & 10 \\ 13 & 20 & 9 \end{pmatrix}, \quad \sum_{j=1}^9 \mu(e_j) = 122 = 3 \cdot 40 + 2.$$

Suppose, for contradiction, that an allocation gives every agent value at least 40. Since $v_X(e_j) \leq \mu(e_j)$ for every agent X and every good e_j , each allocated bundle must have μ -value at least 40. The total μ -value is 122, so each allocated bundle has μ -value at most 42 as well.

There is no one-good bundle with μ -value at least 40. If some allocated bundle has two goods, then it must be a pair with μ -value between 40 and 42. The unique such pair is

$$p = \{e_2, e_4\}, \quad \mu(p) = 16 + 26 = 42.$$

After removing p , the remaining total μ -value is 80, so the other two bundles must both have μ -value exactly 40. The bundle containing e_3 must add goods of total 17 to $\mu(e_3) = 23$. Among the remaining goods, the unique way to obtain total 17 is $\mu(e_5) + \mu(e_7) = 4 + 13$. Hence the only possible partition in the two-good case is

$$p = \{e_2, e_4\}, \quad d = \{e_3, e_5, e_7\}, \quad q = \{e_1, e_6, e_8, e_9\}.$$

The agents' values for these bundles are

	p	d	q
R	42	39	39
C	42	39	39
U	40	40	40

Thus both R and C would need the same bundle p , impossible.

It remains to consider the case in which every allocated bundle has at least three goods. Since there are exactly nine goods and three agents, every bundle has exactly three goods. The three-good bundle containing e_9 must have μ -value between 40 and 42. Since $\mu(e_9) = 9$, the other two goods in this triple must have total value between 31 and 33. The only such pairs are $\{e_7, e_8\}$ and $\{e_3, e_6\}$. Hence the bundle containing e_9 is one of

$$r_3 = \{e_7, e_8, e_9\}, \quad c_3 = \{e_3, e_6, e_9\},$$

and both have μ -value 42.

The remaining two bundles must then have μ -value exactly 40. Consider the remaining bundle containing e_4 , whose μ -value is 26. To reach total 40 with three goods, the other two goods must have total 14. The only possibilities are

$$\{e_5, e_6\} \quad \text{and} \quad \{e_1, e_7\},$$

which give respectively

$$r_2 = \{e_4, e_5, e_6\}, \quad c_1 = \{e_1, e_4, e_7\}.$$

If the e_9 -bundle is r_3 , then e_7 is unavailable, so the e_4 -bundle must be r_2 , and the third bundle is forced to be

$$r_1 = \{e_1, e_2, e_3\}.$$

This is the row partition. Its values are

	r_1	r_2	r_3
R	40	40	40
C	39	39	42
U	39	39	42

so both C and U would need r_3 , impossible.

If the e_9 -bundle is c_3 , then e_6 is unavailable, so the e_4 -bundle must be c_1 , and the third bundle is forced to be

$$c_2 = \{e_2, e_5, e_8\}.$$

This is the column partition. Its values are

	c_1	c_2	c_3
R	39	39	42
C	40	40	40
U	39	39	42

so both R and U would need c_3 , impossible.

All cases lead to contradiction. Therefore no allocation gives all three agents value at least 40. $\square$

6 Consequence for the approximation constant

By Lemma 1, each agent's APS is 40. By Lemma 2, every allocation gives some agent value at most 39. Hence, in this instance, every allocation violates the guarantee

$$v_i(A_i) > \frac{39}{40} \text{APS}_i$$

for at least one agent. Therefore

$$\alpha^* \leq \frac{39}{40}.$$

The bound is tight for this particular instance: assigning

$$A_R = \{e_1, e_2, e_3, e_5\}, \quad A_C = \{e_4, e_7\}, \quad A_U = \{e_6, e_8, e_9\}$$

gives utilities

$$(v_R(A_R), v_C(A_C), v_U(A_U)) = (44, 39, 39).$$

Thus the optimum minimum utility in the instance is exactly 39.

7 What remains open

The argument above is deliberately modest. It proves a direct APS-level obstruction at ratio $39/40$, and so it verifies the upper-bound side of the interval

$$\frac{3}{5} \leq \alpha^* \leq \frac{39}{40}.$$

It does not prove that $39/40$ is attainable in every additive instance. It also does not prove that the $3/5$ lower bound is tight. Therefore the optimal universal approximation constant for APS allocations with additive valuations remains undetermined.

A Exact finite checker

The following script verifies the finite assertions used in the proof. It uses only integer arithmetic and enumerates all 2^9 bundles and all 3^9 allocations. The assignment labels 0, 1, 2 denote agents R, C, U , respectively.

```
from itertools import product

vals = {
    'R': [1,16,23,26,4,10,12,19,9],
    'C': [1,16,22,26,4,9,13,20,9],
    'U': [1,15,23,25,4,10,13,20,9],
}
prices = {
    'R': ([0,3,4,5,1,2,2,4,2], 23),
    'C': ([1,9,12,14,2,5,7,11,5], 66),
    'U': ([1,8,12,13,2,5,7,10,5], 63),
}

# Check APS upper certificates: under the listed prices,
# no affordable bundle has value greater than 40.
for name, v in vals.items():
    q, D = prices[name]
    assert sum(q) == D
    max_affordable = -1
    min_high_weight = None
    for mask in range(1 << 9):
        value = sum(v[j] for j in range(9) if (mask >> j) & 1)
        weight = sum(q[j] for j in range(9) if (mask >> j) & 1)
        if 3 * weight <= D:
            max_affordable = max(max_affordable, value)
        if value >= 41:
            if min_high_weight is None or weight < min_high_weight:
                min_high_weight = weight
    assert max_affordable == 40
    assert 3 * min_high_weight > D

# Check the optimum minimum utility over all allocations.
agent_order = ['R', 'C', 'U']
best_min = -1
best_assignment = None
best_utilities = None
for assignment in product(range(3), repeat=9):
    utilities = [0, 0, 0]
    for item, agent in enumerate(assignment):
        utilities[agent] += vals[agent_order[agent]][item]
    if min(utilities) > best_min:
        best_min = min(utilities)
        best_assignment = assignment
        best_utilities = utilities

assert best_min == 39
assert best_assignment == (0,0,0,1,0,2,1,2,2)
assert best_utilities == [44,39,39]
```

```
print('All_checks_passed.')
```

References

- [1] Moshe Babaioff, Tomer Ezra, and Uriel Feige. Fair-share allocations for agents with arbitrary entitlements. *Mathematics of Operations Research*, 49(4):2180–2211, 2024.
- [2] Uriel Feige, Ariel Sapir, and Laliv Tauber. A tight negative example for MMS fair allocations. In *Web and Internet Economics (WINE 2021)*, Lecture Notes in Computer Science 13112, pages 355–372, 2021.